

Sakayo Toadoum Sari

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PROFILE

NLP researcher working on multilingual evaluation, discourse and causal reasoning, and low-resource African language technologies. PhD candidate at CRIL-CNRS (Université d'Artois). Native speaker of Ngambay, with applied experience in speech and machine translation systems for African languages.

EDUCATION

PhD in Artificial Intelligence

Oct. 2025 – Sep. 2028

CRIL-CNRS, École Doctorale STS, Université d'Artois – Lens, France

- Thesis: *Intertextuality and Causalities: Towards a Semantic Analysis of Discourses*
- Supervised by Pr. L. Saïs, Pr. S. Jabbour, Dr. F. Delorme, Dr. N. Robin (CEPED/IRD) | Funded by Université d'Artois & CRIL

African Master's of Machine Intelligence (AMMI)

Dec. 2021 – Jun. 2023

AIMS – Mbour, Senegal

- Thesis: *Ngambay-French Neural Machine Translation* – supervised by Dr. Angela Fan
- Google & Meta Scholar (full scholarship)

MSc Industrial Mathematics (Data Sciences)

Oct. 2018 – Feb. 2020

AIMS – Limbe, Cameroon

- Thesis: *Hybrid LSTM-Autoregressive Architecture for Time Series Birth Prediction*
- MasterCard Foundation Scholar (full scholarship)

MSc Physics (Electronics, EE, Automation)

Dec. 2017 – Nov. 2019

Université de N'Gaoundéré – Cameroon

- Thesis: *Optimization of Energy Transfer in a Photovoltaic System*

PUBLICATIONS

- Detecting Experiential Intertextuality Across Migration Routes: Beyond Surface Similarity in French Narratives, **SIGDIAL 2026**
- Pretraining Strategies using Monolingual and Parallel Data for Low-Resource MT, **African NLP 2025**
- AfriMTE and AfriCOMET: Empowering COMET for Under-resourced African Languages, **NAACL 2024**
- Ngambay-French Neural Machine Translation (sba-Fr), **NLP4TIA @ RANLP 2023**
- MasakhaNEWS: News Topic Classification for African Languages, **AfricaNLP @ ICLR 2023**
- Error Correction Based DNNs for SA Wildlife-Vehicle Collision Data, **MDPI Mathematics**

In preparation

- RECIT: Relation Extraction with Certified Interpretable Typing for French causal discourse relations
- AfriCOPA: When MT Quality Metrics Lie About Causality. A Diagnostic Probe for African-Language Causal Reasoning

PROFESSIONAL EXPERIENCE

Teaching Assistant

Oct. 2025 – present

Université d'Artois – Lens, France

- Deliver 64 hours of undergraduate algorithmics tutorials (L1 Mathematics/Computer Science) to 80+ students

NLP Engineer

Oct. 2023 – present

Batazia – Rotterdam, Netherlands (Remote)

- Developed and shipped NLP models for 5+ African languages, enabling the company's first multilingual product features
- Built an end-to-end QA framework (automated testing, evaluation metrics, CI integration) for model releases
- Reviewed data handling practices and implemented changes for EU data privacy compliance

Consultant – RCCE

Jul.–Dec. 2025

UNICEF – N'Djamena, Chad

- Delivered the first national mapping of RCCE actors across 10+ regions of Chad, producing an interactive actor database used by UNICEF and government partners

- Designed a mixed-methods research toolkit (surveys, interview guides, focus group protocols) deployed with 50+ grassroots organizations
- Authored a strategic report presented at a multi-stakeholder validation workshop, informing UNICEF's 2026 RCCE programming

Machine Learning Engineer

Apr.–Jul. 2025

CDIAL – Lagos, Nigeria (Remote)

- Trained and evaluated ASR and TTS models for 3+ African languages, testing data augmentation and architecture choices against internal baselines
- Preprocessed 500+ hours of multilingual speech data with reproducible pipelines for onboarding new languages

Independent Consultant – ML Instructor

Mar.–Jun. 2024

Digital Services & GMS Consulting – Douala, Cameroon

- Designed and delivered a 60-hour Machine Learning curriculum to 25+ professionals pursuing a Data Analyst Certificate

Summer Research Intern in NLP

Jun.–Aug. 2023

École Polytechnique Fédérale de Lausanne (EPFL) – Switzerland

- Built predictive models for online polarization detection, reaching 0.82 F1 on a novel benchmark; results contributed to a lab-internal report on content moderation

Lead Trainer in Data

Nov. 2022 – Oct. 2023

Tec4Tchad (Simplon / Univ. de N'Djamena) – Chad

- Trained 40+ Chadian students in Python, Machine Learning, and Git/GitHub over 12 months

Project Lead – ML Intern

Jun.–Sep. 2022

STEM-Away – Santa Clara, USA (Remote)

- Led a 4-person team building a question recommendation engine with Sentence-BERT and Pinecone; promoted to junior consultant

Web Developer & ML Intern

Jul. 2019 – Jan. 2020

Limbe City Council – Limbe, Cameroon

- Designed and deployed a web-based civil status management system (birth, death, marriage certificates), digitizing 10,000+ records

SCHOLARSHIPS & AWARDS

- AMMI full scholarship – Google and Meta Scholar
- AIMS full scholarship – MasterCard Foundation Scholar

PEER REVIEW & SERVICE

- Reviewer for IGI Global (2023, 2024)
- Technical Reviewer, Masakhane African Languages Hub

SKILLS & INTERESTS

Languages French (native) | Ngambay (native) | English (C1) | Portuguese (intermediate)

Technical Python, PyTorch, HuggingFace Transformers, SLURM, AWS (SageMaker, EC2, S3), Docker, Git, $\text{\LaTeX}$

Research Multilingual Evaluation, Benchmark Construction, Discourse Analysis, Intertextuality, Causal Reasoning, Low-Resource NLP, ASR/TTS, Machine Translation, Responsible AI

Interests African language digitization, STEM access for underrepresented communities, community-centered language technologies

REFERENCES

- **Prof. Lakhdar Saïs** – Full Professor, CRIL, Université d'Artois | sais@cril.fr
- **Prof. Said Jabbour** – Full Professor, CRIL, Université d'Artois | jabbour@cril.fr
- **Prof. Marcellin Atemkeng** – RATT, Rhodes University, South Africa | m.atemkeng@ru.ac.za